

Jackson Ferguson

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Education

University of Victoria

Victoria, BC

B.Sc. Major Combined Physics and Astronomy

Graduated Spring 2026

High Performance Coursework: Quantum Computing (100%), Stellar Astrophysics (99%), Probability & Statistics (97%), Computer Assisted Physics (91%), Experimental Physics (88%), Thermodynamics (86%).

Technical Skills

Systems & Hardware:	RP2040/ARM, Signal Processing, Electromechanical Troubleshooting, Oscilloscopes
Software & Scripting:	Python, Bash, TOML/YAML (Config Management), $\text{\LaTeX}$
Data & Statistical Analysis:	NumPy, SciPy, Pandas, Monte Carlo Methods, Matplotlib
Operations & CI/CD:	Git/GitHub Actions, Docker, Environment Orchestration (uv), IaC & Release Automation

Development & Research Projects

Systems Audio Lab: Full-Stack Hardware Integration & Test Platform

Systems Development

- **End-to-End System Integration:** Successfully integrated a four-stage engineering pipeline combining automated inventory logistics, custom power infrastructure, analog circuit fabrication, and digital signal analysis.
- **Test & Measurement Architecture:** Built and calibrated an RP2040-based oscilloscope against the 60 Hz mains frequency; developed an automated Python test suite to extract linear frequency responses (Bode plots).
- **Infrastructure Reliability:** Developed a dedicated linear power supply with physical thermal management to isolate the Device Under Test from the high-frequency switching noise inherent to commercial SMPS units.

Protostar: Development Environment Orchestration System

Python, State Reconciliation, Failure Recovery

- **Environment Orchestration:** Built a declarative orchestration layer for Python environments, coordinating package resolution, Git hooks, Docker, CI/CD, documentation, IDE configuration, and task runners from composable recipes.
- **State Convergence:** Recorded desired state and ownership baselines, then implemented `status/diff/sync` to reconcile live repositories, retract retired managed content, and preserve operator changes.
- **Failure Containment:** Designed a two-phase plan/execute boundary with stale-input checks, mutation journaling, subprocess termination, and byte/permission rollback after failures or interrupts.
- **Operational Interfaces:** Exposed the same engine through Textual review and conflict resolution plus headless JSON/CI interfaces, with explicit POSIX statuses, diagnostics, and versioned template migrations.

Star Ground: Automated Logistics & Tech Ops Platform

Python, Streamlit, Systems Integration

- **Automated Yield Management:** Designed a risk-aware sourcing algorithm that calculates net-need deficits against a stateful inventory model, applying category-specific safety buffers to balance unit economics against infinite stockout costs.
- **High-Observability Architecture:** Implemented a transparent, “loud-failure” User Interface that strictly separates residual noise from logic exceptions, surfacing real-time telemetry and generating diagnostic artifacts.

CI/CD & Release Infrastructure

GitHub Actions, Python, Release Automation

- **Two-Phase Release Orchestration:** Implemented read-only preflight checks for branch, cleanliness, remote synchronization, version metadata, and tag collisions before any release state is mutated.
- **Transactional Publication:** Automated version/lock updates, release commits, annotated tags, and `git push --atomic`, with local rollback when execution fails or is interrupted.
- **Reusable Distribution Pipelines:** Built centralized workflows that wait for PyPI artifacts, resolve dependency trees with `uv`, generate Homebrew resources, and audit updated formulae for caller repositories.

Galaxy Cluster Dark Matter Analysis (ACO 2670)

Astrophysics Research Project

- **Pipeline Development:** Constructed a data pipeline to clean spectroscopic data for 98+ galaxies, utilizing redshift-space distortion analysis to isolate cluster members from the background field.
- **Statistical Modeling:** Applied the Virial Theorem and Gaussian Mixture Models (GMM) to calculate a mass-to-light ratio of $(291 \pm 60) M_{\odot}/L_{\odot}$, providing kinematic evidence of Dark Matter dominance.

Work & Leadership Experience

Co-Lead Event Organizer, Unplugged Events, Victoria, BC

Jan 2025 – Jan 2026

- Managed technical production, signal flow, and venue logistics for live events (100+ attendees).

Server, Stanley Park Brewing, Vancouver, BC

May 2025 – August 2025

Alpine Ski Instructor, Whistler Blackcomb, Whistler, BC

Nov 2018 – Feb 2020

Freelance 3D Modeler, Twisted Time Studios, Remote

May 2015 – Aug 2015

- Designed and created main character models and weapon systems for *Extinction: Archaic Genesis*.